

Process Design for Biodiesel Production from Used Cooking Oil

**ChE62 Materials balances and separation processes Project
Curtis Lee**

Abstract	2
Full Process Description	3
Reaction Selection and Justification	3
Process Simplifications for Modelling	4
Simplified components:	4
Simplified Process step description	5
Design Specification	6
Mass and energy balances	7
Overall Mass and energy balance	10
Cost analysis	11
Modes of failure analysis	12
Works Cited	16

Abstract

This report presents a process design for producing biodiesel from used cooking oil through pretreatment, esterification, transesterification, separation, and methanol recovery. The feedstock is first filtered and dried to remove solids and water. Because used cooking oil contains free fatty acids (FFAs) that can form soap during base-catalyzed reactions, an acid-catalyzed esterification pretreatment is used to convert the FFAs into fatty acid methyl esters (FAMEs).

In order to calculate theoretical material balances, a simplified model is created. It treats the oil as 85% triolein and 15% oleic acid, allowing the process to focus on the major triglyceride and free fatty acid components. The main transesterification step reacts triolein with excess methanol using KOH catalyst to produce fatty acid methyl esters and glycerol.

The process is designed for 3000 L/day of used cooking oil and uses two 500 L CSTRs, a 1000 L gravity decanter, and a 100 L methanol recovery still. Material and energy balances show that waste heat recovery can reduce energy demand from 284.6 kWh/day to about 122 kWh/day. The estimated equipment cost is under \$200,000, while daily raw material, delivery, and energy costs are approximately \$2000.

Full Process Description

Used cooking oil is screened and filtered to remove solids. This is necessary because the oil can contain water, meat scraps, and other contaminants that must be removed before use as biodiesel feedstock.

As water and free fatty acids interfere with base-catalyzed biodiesel production and can undergo saponification (soap formation), so the oil is heated and dried to remove excess dissolved water.

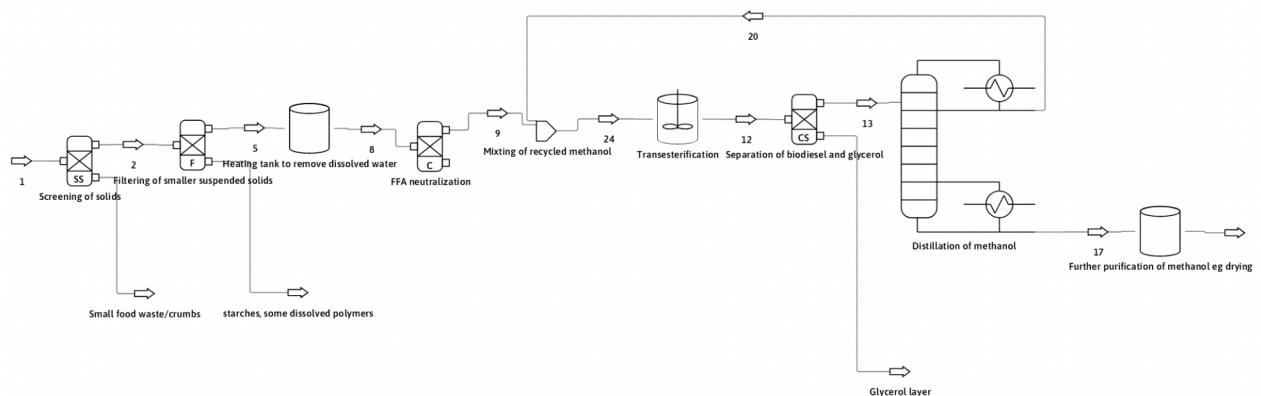
The remaining free fatty acids can undergo saponification (making soap) with the base catalysts in the transesterification step so they must be neutralized. The oil must then be tested for free fatty acid content via titration. If the acid level is high, the oil goes through an acid esterification pretreatment using methanol and sulfuric acid. This converts free fatty acids into methyl esters and lowers the acidity before the main base-catalyzed step.

After pretreatment, the main unit operation is a CSTR. The dried oil reacts with excess methanol in the presence of a strong base such as KOH or NaOH. Transesterification converts fats into biodiesel and glycerin.

The solution is sent to a decanter. Two liquid layers form, a lighter biodiesel-rich phase and a heavier glycerol-rich phase. The glycerol layer contains glycerol, excess methanol, catalyst, soaps, and dissolved impurities. The biodiesel layer contains FAME, leftover methanol, trace catalyst, soaps, and unreacted glycerides.

To recycle the methanol and reduce cost and waste, we can perform flash evaporation or distillation. The biodiesel is then washed or dry-polished to remove residual catalyst, soaps, monoglycerides, glycerol, and salts; washing/filtration after transesterification is used to remove impurities from raw biodiesel.

Finally, the biodiesel is dried and filtered to produce finished biodiesel. The glycerol coproduct is neutralized and purified if it is to be sold or reused.

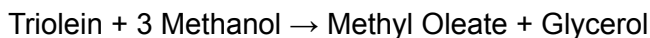


Reaction Selection and Justification

The key reaction is transesterification of triglycerides with methanol:



We can use Triolein as the main triglyceride in cooking oil



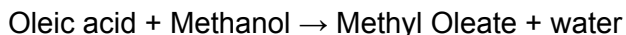
This reaction is selected because it directly converts the triglycerides in used cooking oil into Fatty acid methyl esters (FAME) biodiesel, the desired liquid fuel product. Methanol is commonly used, and NaOH or KOH is commonly used as the base catalyst.

A practical reaction condition is approximately 6:1 methanol:oil molar ratio, about 1 wt% KOH or NaOH catalyst, strong mixing, and moderate heating near methanol's boiling point. Excess methanol is used because the reaction is reversible and excess alcohol drives conversion toward biodiesel. This drives a conversion of nearly 95%.

Due to the acidic nature of used cooking oil which can react with the base catalyst, the pretreatment reaction is acid-catalyzed esterification:



We can use Oleic acid as our main example of free fatty acids:



This step is justified because it converts free fatty acids into biodiesel-range methyl esters instead of allowing them to form soap (undergo saponification) during the base-catalyzed step. The H₂SO₄ can then be neutralized with excess base on the transesterification step. This step also has a near 100% conversion rate due to the strong acid base reaction.

The saponification reaction with oleic acid:



Overall, this pathway is appropriate because used cooking oil is inexpensive, recycled, and already rich in triglycerides. Transesterification is the standard process for converting these fats/oils into biodiesel and glycerol.

Process Simplifications for Modelling

Simplified components

The approximate composition of used cooking oil is [5] :

- Triglycerides: 80% - 90%
- Free Fatty Acids (FFA): 5% - 15%
- Moisture: $\leq 1\%$
- Impurities: $\leq 0.5\%$

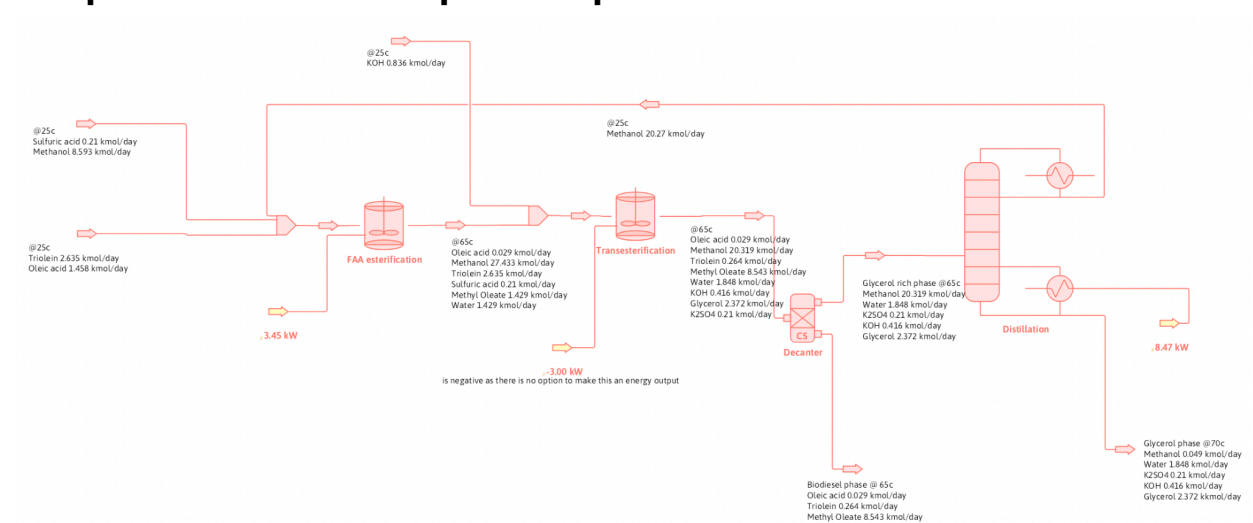
Typical Fatty Acid Profile:

- Oleic Acid (C18:1): 30% - 45%
- Linoleic Acid (C18:2): 20% - 30%
- Palmitic Acid (C16:0): 10% - 15%
- Stearic Acid (C18:0): 5% - 10%
- Other Fatty Acids: 5% - 10%

As the amounts of moisture and impurities are very low, we can model the used cooking oil to compose of only Triglycerides and Free Fatty Acids. The FFAs (Free fatty acids) primarily compose of Oleic acid, and the Triglycerides compose primarily of Triolein [8]. This allows us to model used cooking oil as a composition of 85% Triolein and 15% Oleic acid.

This simplification reduces the number of our components to Oleic acid and Triolein (in our feedstock), sulfuric acid (catalyst for the FFA esterification), methanol (Reactant with triolein or FFAs), and Potassium Hydroxide (catalyst for transesterification).

Simplified Process step description



Step 1: FFA esterification

Esterification of Oleic acid. Using a 19.8:1 methanol:FFA molar ratio, and a 5g Sulfuric acid to 1g FFA. At 65c, and after 2 hours, the reaction conversion is around 98% reducing the FFA concentration to 0.5 wt% [10].

Step 2: Neutralization of Sulfuric acid catalyst and transesterification of the Triolein

The mixture of methyl-esters, Oleic acid and sulfuric acid catalyst is transferred to a separate CSTR. And 1 wt% KOH catalyst is added, since there is excess methanol from step 1, above 6:1 methanol:oil ratio, no additional methanol needs to be added. Separately, the KOH is replenished constantly to offset the neutralization reaction with sulfuric acid, as this is a strong acid-base reaction, the conversion of this reaction is 100%. The transesterification reaction occurs at >60c and the reaction proceeds for 90 minutes, and this reaction occurs with 90% conversion [11]. As the amount of water formed is low (still below 1 wt%), we can assume that the hydrolysis reaction of triolein is negligible.

Step 3: Separation of biodiesel from glycerol layer

Separation of Glycerol rich and biodiesel rich stages can be achieved via gravity as they have significantly different densities. This process can occur in a tank/decanter around 4 hours [9].

Step 4: Distillation of Glycerol layer to recycle methanol

The glycerol layer is known as the waste layer in transesterification. It contains most of the waste material such as the catalyst, methanol and water. Due to the low boiling point of methanol, flash distillation of the glycerol mixture can be performed to remove the excess methanol at a relatively low temperature. This is usually done around 70c, and reduces the remaining methanol to 0.5 wt% in the glycerol layer within an hour [9].

Design Specification

Step 1 takes in 3000 L/day of used cooking oil, 1168 L/day of methanol and 11 L/day of sulfuric acid. And the residence time is 2 hours. This gives an hourly input flow rate of 174.1 L/hr, so at any given time there will be 348.2 L within the CSTR. To account for mixing, and level control, a 500 L insulated CSTR with heating is selected.

Step 2 takes in approximately 4200 L/day of mixture from step 1, which gives an hourly input flow rate of 175 L/hr. The residence time is 90 minutes, so the reactor needs to hold 328 L. For the same reasons as in step 1, a 500 L jacketed CSTR with cooling is selected.

Step 3 takes in about 4200 L/day of mixture from step 2, and has a residence time of at least 4 hours. So the reactor must hold at least 700 L, so a 1000 L gravity decanter/settling tank is selected.

Step 4 takes in only the glycerol rich layer, and that is approximately 1050 L/day, and 43.8 L/hr. A small flash still requires 1 hour per batch, and requires significant vapor space (around 100% of liquid), so a 100L still with condenser is selected. The methanol extracted will be fed back into step 1 as a recycle stream.

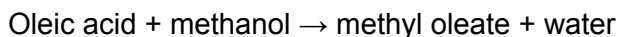
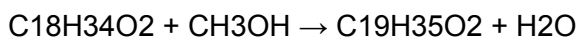
Specific flow rates and composition of each stream are calculated below in the Mass and energy balances section.

Mass and energy balances

The process is designed for 3000 L/day used cooking oil feed. Using the data sheet, the used cooking oil has an average density of 0.915 kg/L. This means that 2745 kg/day of used cooking oil is processed.

Step 1 Requires 0.05g H₂SO₄ per 1g of FFA (Oleic acid), so the sulfuric acid loading is 20.59 kg/day.

FAA esterification reaction:



This reaction occurs at 98% conversion and the reaction has a $\Delta H = -10.31$ kJ/ mol oleic acid reacted. Using a 19.8:1 methanol:FFA molar ratio, and a 5g Sulfuric acid to 1g FFA. FFA concentration is reduced to 0.5 wt%

For 98% conversion:

Input stream	Molar flow rate (kmol/day)	Output stream	Molar flow rate (kmol/day)
Oleic acid	1.458	Oleic acid	0.029
Methanol	28.862	Methanol	27.433
Triolein	2.635	Triolein	2.635
Sulfuric acid	0.210	Sulfuric acid	0.210
		Methyl oleate	1.429
		water	1.429

Energy balance

The initial feed is heated from 25c → 65c, $\Delta 40c$

Taking heat capacities of Triolein/oleic acid : 2 kJ/kgK, Methanol: 2.5 kJ/kgK, Sulfuric acid: 1.4 kJ/kgK.

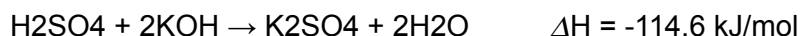
The heating required = 313.22 MJ/day

The heat released from esterification = 14.73 MJ/day.

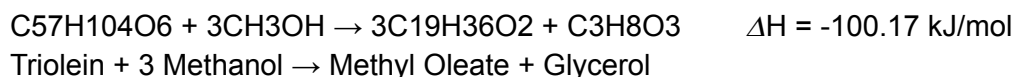
So the net heat required for Step 1 to maintain a temperature of 65c is 298.49 MJ/day or 3454.745 watts. Assuming no heat is lost from the reactor.

Step 2 requires 1:2 H₂SO₄:KOH molar ratio to neutralize H₂SO₄ with KOH at 100% conversion. Then a transesterification reaction with 6:1 methanol:oil molar ratio at 60c and 1 wt% KOH with 90% Triolein conversion. As there is more than enough (10.3:1) methanol leftover from step 1, no additional methanol needs to be added. This excess of methanol will increase the conversion, but make separation take longer. This difference will be neglected in this simplification as the effects are not significant [12].

Neutralization reaction:



Transesterification reaction:



Amount of KOH needed is

As a catalyst (1 wt% triolein) = 23.33kg/day = 0.416 kmol/day

For neutralization (1:2) = 0.420 kmol/day

For 90% conversion, and note that there is enough methanol from step 1, at a 10.3:1 ratio, this is higher than the ideal of 6:1, this doesn't affect the conversion, only that separation is worse as methanol helps the layers emulsify.

Input stream	Molar flow rate (kmol/day)	Output stream	Molar flow rate (kmol/day)
Oleic acid	0.029	Oleic acid	0.029
Methanol	27.433	Methanol	20.319
Triolein	2.635	Triolein	0.264
Sulfuric acid	0.210	Sulfuric acid	0
Methyl oleate	1.429	Methyl oleate	8.543
water	1.429	water	1.848
KOH	0.836	KOH	0.416
		Glycerol	2.372
		K ₂ SO ₄	0.210

Energy balance

The step 2 feed enters at 65c, so the mixture does not need to be heated.

There is no energy required for this step other than the CSTR stirring mechanism.

However, the reaction is exothermic so the output flow will be at a different temperature. The heat released from neutralization is $0.21(1000)(114.6) = 24.06$ MJ/day. And the heat released from transesterification is $2.372(1000)(100.17) = 237.56$ MJ/day. The heat required to raise KOH from 25c to 65c is 2.44 MJ/day

Using the combined heat capacity of each component, the reaction output temperature will be at 95c. This will have to be cooled back down to 60-65c before step 3 , though the heat/energy can be recycled for either step 1 or step 4.

The energy released (assuming no loss in extraction) would be 259.18 MJ/day or 2999.77W

Step 3 allows for the reaction mixture to separate. As specific values are difficult to find under previous assumptions, and it is stated in literature that the glycerol layer contains most of the waste materials after biodiesel production [4][9][10]. For this simplified material balance, we can assume no methanol or water dissolves in the biodiesel rich phase. And since nearly all of KOH and K₂SO₄ is dissolved in water or methanol, it is contained in the glycerol rich phase.

Input stream	Molar flow rate (kmol/day)	Output stream (biodiesel)	Molar flow rate (kmol/day)	Output stream (glycerol)	Molar flow rate (kmol/day)
Oleic acid	0.029	Oleic acid	0.029	Methanol	20.319
Methanol	20.319	Triolein	0.264	Water	1.848
Triolein	0.264	Methyl oleate	8.543	K ₂ SO ₄	0.210
Sulfuric acid	0			KOH	0.416
Methyl oleate	8.543			Glycerol	2.372
water	1.848				
KOH	0.416				
Glycerol	2.372				
K ₂ SO ₄	0.210				

Step 4 distills methanol out of the glycerol rich phase until 0.5 wt% remains in the layer at 70c. Assuming there is no water or glycerol that evaporates, to remain with 0.5 wt% methanol, that is 0.049 kmol/day of methanol remaining in the glycerol layer.

Input stream	Molar flow rate (kmol/day)	Output stream	Molar flow rate (kmol/day)	Output stream (distillate)	Molar flow rate (kmol/day)
Methanol	20.319	Methanol	0.049	Methanol	20.270
Water	1.848	Water	1.848		
K ₂ SO ₄	0.210	K ₂ SO ₄	0.210		
KOH	0.416	KOH	0.416		
Glycerol	2.372	Glycerol	2.372		

Energy balance:

Heating from 65 to 70c = 11.79 MJ/day or 135.46 W

Methanol vaporization (20.270 kmol/day) = 720.20 MJ/day or 8335 W

Overall Mass and energy balance

Mass balance

Accounting for the recycle stream, only the methanol lost or reacted has to be continuously replenished.

Input streams	Molar flow rate (kmol/day)	Output stream (biodiesel)	Molar flow rate (kmol/day)	Output stream (distillate, glycerol layer)	Molar flow rate (kmol/day)
Triolein	2.635	Oleic acid	0.029	Methanol	0.049
Oleic acid	1.458	Triolein	0.264	Water	1.848
Methanol	8.593	Methyl oleate	8.543	K ₂ SO ₄	0.210
Sulfuric acid	0.210			KOH	0.416
KOH	0.836			Glycerol	2.372

Energy balance (not accounting for equipment operation)

Assuming energy lost in cooling and condensing is lost, heating of the components over steps 1-4 requires 1024.7 MJ/day or 284.6 kWh/day. And the energy that is lost from cooling and condensing is 1046.65 MJ/day or 290.7 kWh/day.

If we recover the heat from cooling step 2 and the condensing process in step 4, we can greatly offset the energy required to run the plant. According to reports from ASHRAE and the California Energy Commission, typical heat recovery exchangers have effectiveness values of 50-70%.

Taking an effectiveness of 60%, this decreases the energy required for the process to 441MJ/day, or 122kWh/day.

Cost analysis

Initial costs (excluding land/rent etc)

Equipment	Cost estimate (\$)	Source
500 L jacketed CSTR	50k	https://www.acrossinternational.com/shop-prod

500 L jacketed CSTR	50k	uct-by-categories/stainless-glass-reactors/stainless-steel-reactors/stainless-500l/ai-dual-jacketed-500l-316l-grade-sst-reactor-a.html?srltid=AfmBOopAARdFU-FFaigYOhKqVus1U8NohuY8sRDxcmYs6wsgskTZHSyivtc
1000 L gravity decanter	30k	https://www.alibaba.com/product-detail/Three-Phase-Waste-Oil-Treatment-Centrifuge_1600105260706.html?spm=a2700.7724857.0.0.98e22041NZsRdL
100 L distilling apparatus	20k	https://www.alibaba.com/product-detail/Laboratory-Molecular-Distillation-Equipment-For-Small_1601709348562.html?spm=a2700.galleryofferlist.p_offer.d_title.1adf13a0UxQt4y&priceId=27a89da34e584632a6024eaf83f5f882
Pumps, piping etc	5-10k	estimations
Energy recovery	0.5-1k	https://altheatsupply.com/products/stainless-steel-tube-and-shell-heat-exchanger-100k-btu

Estimates on equipment costs, not including installation prices.
This brings the initial equipment cost to just under \$200,000.

Item	Amount needed (/day)	Unit cost (\$)	Daily cost (\$)	Source
Used cooking oil	3000L	0.495/gallon	392.30	Local source Price estimate from usda https://mymarketnews.ams.usda.gov/viewReport/3510
Methanol	8593 mol or 275.34 kg	1.48/kg	407.50	Methanex
Sulfuric acid	210mol or 20.597 kg	0.1165/kg	2.40	IMARC group
KOH	836 mol or 46.91 kg	0.95/kg	44.28	IMARC group
Electricity (with recovery)	122 kWh	0.33/kWh	40.26	California public utilities
Electricity (without recovery)	284.6 kWh	0.33/kWh	93.918	California public utilities

With waste heat recovery, the daily raw-material and energy cost is \$980.66.

This does not account for transport and delivery costs, an estimate for each UCO delivery each day of 3000 L costs in the range of 700-800 dollars [14]. This gives an estimated daily raw-material and energy cost of \$2000. The key costs every day are the delivery of the used cooking oil and the price of methanol.

Modes of failure analysis

Possible operating errors in this biodiesel process mainly come from variation in feedstock quality, reaction control, separation problems, and methanol recovery. We can address them by adding testing, control loops, safety margins, and recycle/purge checks to the process design.

1. Variation in used cooking oil composition

Used cooking oil may not always match the simplified design assumption of 85% triolein and 15% oleic acid. The actual FFA, water, and impurity content can vary significantly depending on the oil source. Since the process assumes FFAs are reduced by esterification before base transesterification, a higher-than-expected FFA level could consume extra KOH later and form soaps.

To address this, feedstock pretreatment and a testing step before the main process can be included. Each batch or daily feed should be tested for FFA content by titration and for water content. The sulfuric acid and methanol feed rates in Step 1 should then be adjusted based on the measured FFA level rather than assuming a constant 15% FFA.

2. Incorrect acid or base dosing

Step 1 uses sulfuric acid for FFA esterification, while Step 2 uses KOH both to neutralize sulfuric acid and catalyze transesterification. If too little KOH is added, remaining acid will lower the base catalyst activity and reduce conversion. If too much KOH is added, soap formation and contamination of the glycerol phase may increase.

To address this, dosing pumps and pH meters can be used. They can continuously measure pH levels, adding KOH gradually to neutralize the acid.

3. Temperature control errors

Step 1 is designed around 65°C, while Step 2 operates above 60°C and is exothermic. The report estimates that Step 2 could rise to around 95°C if heat is not removed. That is a major operating risk because methanol is volatile, higher temperatures increase vapor pressure, and excessive heat could be dangerous.

To address this, Step 2 should use a jacketed CSTR with active cooling, and temperature alarms. A larger CSTR can also be used in case of boil over. Since Step 2 releases useful heat, the heat recovery exchanger should be designed so that it cannot overheat Step 1 or Step 4.

4. Incomplete mixing or poor residence time control

Both reaction steps depend on sufficient residence time: Step 1 is designed for 2 hours, and Step 2 for 90 minutes. Poor mixing, incorrect flow rate, or reactor level control errors could cause incomplete esterification or transesterification.

To address this, level sensors, flow meters, and agitation monitoring on both CSTRs can be attached. The reactors were already oversized to 500 L for process margin, which helps, but the design should also include sampling ports to check FFA conversion after Step 1 and biodiesel conversion after Step 2.

5. Methanol evaporation and flammability hazard

Methanol is used in large excess, recycled from the glycerol layer, and heated near its boiling point. Methanol loss, vapor leaks, or poor condenser performance could create a fire and toxicity hazard. Step 4 also distills methanol at about 70°C, so vapor handling is important.

To address this, the methanol system should be designed as a closed system with condensers, vapor recovery, pressure relief, ventilation, explosion-proof electrical equipment, and methanol leak detection. The recycled methanol stream should also be tested so that water or glycerol contamination does not build up over time.

6. Poor biodiesel/glycerol separation

The design relies on gravity separation in a decanter for about 4 hours. However, excess methanol, soap, water, and fine impurities can stabilize emulsions and make the glycerol and biodiesel layers separate poorly. The report itself notes that excess methanol can make separation worse.

To address this, the decanter should be oversized, include a clear interface-level sensor, and allow longer settling time when needed. If separation is stubborn, the design could include gentle heating, water washing, salt/brine addition, centrifugation, or acid neutralization to break emulsions.

Works Cited

- [1] U.S. Department of Energy. (n.d.). *Biodiesel production*. Alternative Fuels Data Center.
<https://afdc.energy.gov/fuels/biodiesel-production>
- [2] National Renewable Energy Laboratory. (2009). *Biodiesel handling and use guide* (4th ed.).
<https://docs.nrl.gov/docs/fy17osti/66521.pdf>
- [3] California Advanced Biofuels Alliance. (n.d.). *Frequently asked questions*.
<https://www.caadvancedbiofuelsalliance.org/faq>
- [4] Rachmanto, T. A. (2015). *Design of biodiesel production process from waste cooking oil* (Master's thesis, Liverpool John Moores University).
<https://researchonline.ljmu.ac.uk/id/eprint/4337/>
- [5] <https://alexchem-ltd.com/>
- [6] Chai, Ming, et al. "Esterification Pretreatment of Free Fatty Acid in Biodiesel Production, from Laboratory to Industry." *Fuel Processing Technology*, vol. 125, Sept. 2014, pp. 106–113, digitalcommons.unl.edu/cgi/viewcontent.cgi?article=1212&context=usepapapers,
<https://doi.org/10.1016/j.fuproc.2014.03.025>.
- [7] Erchamo, Yeshimebet Simeon et al. "Improved biodiesel production from waste cooking oil with mixed methanol-ethanol using enhanced eggshell-derived CaO nano-catalyst." *Scientific reports* vol. 11,1 6708. 23 Mar. 2021, doi:10.1038/s41598-021-86062-z
- [8] Peters, Morenike A et al. "Subcritical Water Hydrolysis of Fresh and Waste Cooking Oils to Fatty Acids Followed by Esterification to Fatty Acid Methyl Esters: Detailed Characterization of Feedstocks and Products." *ACS omega* vol. 7,50 46870-46883. 5 Dec. 2022, doi:10.1021/acsomega.2c05972

[9] Burgess Clifford, Caroline. "8.3 Various Processes Used to Make Biodiesel." *EGEE 439: Alternative Fuels from Biomass Sources*, College of Earth and Mineral Sciences, The Pennsylvania State University, courses.ems.psu.edu/egEE439/node/685. Accessed 1 May 2026.

[10] Colin J. Stacy, Cory A. Melick, Richard A. Cairncross,
Esterification of free fatty acids to fatty acid alkyl esters in a bubble column reactor for use as biodiesel,

[11] Falah Fahed Banihani. (2017). Transesterification and Production of Biodiesel from Waste Cooking Oil: Effect of Operation Variables on Fuel Properties. *American Journal of Chemical Engineering*, 4(6), 154-160. <https://doi.org/10.11648/j.ajche.20160406.13>

[12] Kartikkumar Thakkar, Surendra Singh Kachhwaha, Pravin Kodgire,
Multi-response optimization of transesterification reaction for biodiesel production from castor oil assisted by hydrodynamic cavitation, *Fuel*, Volume 308, 2022,

[13]
https://www.alibaba.com/product-detail/Competitive-Price-100L-200L-500L-700L_1601238828331.html?spm=a2700.7724857.0.0.4b2b4651by5221

[14] <https://www.sandsbrokerageinc.com/blog/ftl-shipping-cost-usa-2026/>